Ex-4

Name: P V Gridhar
RegNo: 19722600zz

Title:

Construct a CPU Scheduling Program in C using the Shortest Job First (SJF) Technique.

SOURCE CODE:

```
#include <stdio.h>
int main() {
    int n,i,j,bt[20],wt[20],tat[20],p[20],t;
    float aw=0,at=0;
    printf("Enter number of processes:");
    scanf("%d",&n);
    for(i=0;i<n;i++){
        p[i]=i+1;
        printf("Burst time P%d:",i+1);
        scanf("%d",&bt[i]);
    }
    for(i=0;i<n-1;i++)
        for(j=i+1;j<n;j++)
            if(bt[i]>bt[j]){
                t=bt[i];bt[i]=bt[j];
bt[j]=t;
                t=p[i];p[i]=p[j];p[j]=t;
            }
    for(i=0;i<n;i++){
        wt[i]=(i==0)?0:wt[i-1]+bt[i-1];
```

```
    tat[i] = wt[i] + bt[i];
    aw += wt[i];
    at += tat[i];
    printf("P%d\t%d\t%d\t%d\n", p[i], bt[i],
           wt[i], tat[i]);
  }
  printf("Average WT = %.2f\n", aw/n);
  printf("Average TAT = %.2f\n", at/n);
  return 0;
}
```

Sample Output:

P_2	3	0	3
P_1	5	3	8
P_3	8	8	16

Average WT = 3.67
Average TAT = 9.00